Auteur: Stan Van Campenhout
Studierichting: Informatica
Oefeningenreeks 4A nr. 23.2

$$\int x \operatorname{Bgtan} x dx = \frac{1}{2} \int \operatorname{Bgtan} x d(x^2)$$
$$\frac{x^2}{2} \operatorname{Bgtan} x - \frac{1}{2} \int x^2 d\left(\operatorname{Bgtan} x\right) = \frac{x^2}{2} \operatorname{Bgtan} x - \frac{1}{2} \int \frac{x^2}{1+x^2} dx$$
$$= \frac{x^2}{2} \operatorname{Bgtan} x - \frac{1}{2} \int \frac{1+x^2-1}{1+x^2} dx$$
$$= \frac{x^2}{2} \operatorname{Bgtan} x - \frac{1}{2} \int \frac{1+x^2}{1+x^2} dx + \frac{1}{2} \int \frac{1}{1+x^2} dx$$
$$= \frac{x^2}{2} \operatorname{Bgtan} x - \frac{1}{2} x + \frac{1}{2} \operatorname{Bgtan} x + c = \frac{x^2+1}{2} \operatorname{Bgtan} x - \frac{1}{2} x + c$$

## References